

## 1.3.5 – 1.3.6 TCP/IP Model vs OSI Model

ٹی سی پی / آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔

CS XI

Chapter 1: Computer Systems

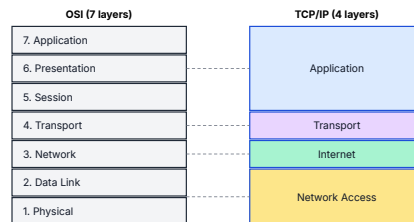
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#### LEARN IT

The OSI model is mainly a **reference** model used for teaching and design. The **TCP/IP model** is the **practical model used on the real Internet**. It condenses the 7 OSI layers into just **4 layers**.

| TCP/IP model (4 layers) | Corresponds to OSI layers                    |
|-------------------------|----------------------------------------------|
| 1. Application          | Application, Presentation, Session (7, 6, 5) |
| 2. Transport            | Transport (4)                                |
| 3. Internet             | Network (3)                                  |
| 4. Network Access       | Data Link, Physical (2, 1)                   |



How the 7 OSI layers map onto the 4 TCP/IP layers.

| Point            | OSI model                                             | TCP/IP model                                  |
|------------------|-------------------------------------------------------|-----------------------------------------------|
| Number of layers | 7                                                     | 4                                             |
| Use              | Reference / teaching model                            | Practical model used on the real Internet     |
| Top layers       | Separate Application, Presentation and Session layers | All three combined into one Application layer |
| Bottom layers    | Separate Data Link and Physical layers                | Combined into one Network Access layer        |

#### WORKED EXAMPLE

**Where does each protocol/device go in TCP/IP?**

HTTP (OSI layer 7) → TCP/IP **Application**.

SSL/TLS (OSI layer 6) → TCP/IP **Application** (layers 7, 6, 5 are merged).

UDP (OSI layer 4) → TCP/IP **Transport**.

IP and routers (OSI layer 3) → TCP/IP **Internet**.

Ethernet, switches, cables (OSI layers 2 and 1) → TCP/IP **Network Access**.

#### REMEMBER IT

Count it as **3-1-1-2**: the TCP/IP Application layer takes 3 OSI layers, Transport takes 1, Internet takes 1, Network Access takes 2. Total  $3+1+1+2 = 7$ .

#### COMMON MISTAKE

The OSI "Network" layer (3) matches the TCP/IP "**Internet**" layer, not the TCP/IP "Network Access" layer. The similar names confuse many students.

اردو میں سمجھیں

ٹی سی پی / آئی پی (TCP/IP) ماڈل حقیقی انٹرنیٹ پر استعمال ہونے والا عملی ماڈل ہے اور اس کی صرف چار تہیں ہیں۔ اس کی ایپلی کیشن تہ OSI کی 7، 6 اور 5 تہوں کے برابر ہے، ٹرانسپورٹ تہ 4 کے، انٹرنیٹ تہ 3 کے، اور نیٹ ورک ایکسیس تہ 2 اور 1 کے برابر ہے۔

#### CHECK YOURSELF

1. How many layers does the TCP/IP model have?
2. Which TCP/IP layer corresponds to OSI layers 2 and 1?
3. Which TCP/IP layer contains the OSI Session layer?

Answers: 1. Four. 2. Network Access. 3. Application.

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Next lecture: Chapter exam (in the chapter lecture PDF)